SWINBURNE VIETNAM

HO CHI MINH CAMPUS

COS30018 - COS30018-Intelligent Systems

Oral Defense week 8

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1. In stock prediction, why is Mean Squared Error (MSE) or Mean Absolute Error (MAE) often preferred over simple "Accuracy" used in classification?

Stock prediction is a regression problem: the output is a continuous price value, not a discrete class label.

- Accuracy only makes sense for classification ("correct" vs "incorrect" labels).
- In regression, we care about how far the prediction is from the real price.
    - MAE tells the average absolute error in price units (easy to interpret: "off by about 2.8 dollars/points").
    - MSE squares the errors, so it penalizes large mistakes more (useful when big errors are especially bad).
    - RMSE is sqrt(MSE), returning to price units but still emphasizes big errors.

In my experiments I report MAE/MSE/RMSE because they directly measure prediction quality for continuous prices.

2. How does the choice of "Batch Size" and "Number of Epochs" impact the convergence of your stock price model?

They directly affect how the model learns and whether it underfits/overfits.

Batch size

- Smaller batch (e.g., 16) updates weights more often → noisier gradients → can generalize better but is slower per epoch.
- Larger batch (e.g., 64) gives smoother gradients → faster per epoch, but can converge to a different minimum and sometimes generalize worse.

Epochs

- More epochs usually improves fit at the start (loss decreases), but after a point may cause overfitting (training improves but test error stops improving or worsens).
- Too few epochs can lead to underfitting (model hasn't learned patterns yet).

In my runs, changing from (epochs=25, batch=64) to (epochs=32, batch=50) changed model performance noticeably:

- LSTM improved a lot (MAE went from 5.77 → 3.83),
- GRU improved slightly (2.89 → 2.78),
- RNN slightly worsened in MAE (4.28 → 4.62).

So convergence behavior depends on both batch/epochs and the architecture.

3. Compare the performance of an LSTM network vs. a GRU network for your specific stock. Which one was faster to train, and which was more accurate?

Setup (fair comparison):

- Same stock: CBA.AX
- Same features: Open, High, Low, Close, Volume
- Same lookback: 60
- Same architecture size: 3 layers × 50 units, dropout 0.2
- Same evaluation: test MAE/MSE/RMSE
- Two training configs tested.

Accuracy (lower MAE/MSE/RMSE is better)

GRU was more accurate in both configs:

- Epochs 25 / Batch 64
    - GRU MAE 2.891 vs LSTM MAE 5.770
- Epochs 32 / Batch 50
    - GRU MAE 2.782 vs LSTM MAE 3.826

So, in my experiment, GRU consistently predicted better than LSTM on the test set.

Training speed

In my measured training times, LSTM was faster than GRU for both configs:

- Epochs 25 / Batch 64: LSTM 14.24s vs GRU 16.74s
- Epochs 32 / Batch 50: LSTM 19.41s vs GRU 21.59s

So for my environment and setup:

- Faster to train: LSTM
- More accurate: GRU